DATA MANAGEMENT - Project proposal

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Type of project:

[DW] Select a set of data sources, integrate them by means of ETL operations (you can either use an integration tool or write simple scripts to solve common ETL problems), define the DFM model for your data, define and populate the corresponding star schema or snowflake schema, by using a relational DB (e.g., Postgres).

Dataset:
- ***European Football Games:*** https://www.kaggle.com/datasets/waterchiller/european-football-games

Data about matches in the top 5 leagues (Serie A, Premier League, La Liga, Ligue 1, Bundesliga) from 2004/2005 to 2018/2019. I'm interested in **Away Coach**, **Away Goals**, **Away Name**, **Date**, **Home Coach**, **Home Goals**, **Home Name**, **League**, **Referee**, **Season**, **Stadium**, **Visitor Count**.
- ***Football Data: Competitions, Clubs, Players:***

https://www.kaggle.com/datasets/thedevastator/football-data-competitions-clubs-players-statist

Data about all the European Football Competitions. I'm interested in **Name**, **Type**, **Country Name**.
- ***Top 5 European football leagues teams stats:***

https://www.kaggle.com/datasets/jehanbhathena/big-5-european-football-leagues-stats

Statistics about the top 5 European Football Leagues from 2010/2011 to 2020/2021. I'm interested in **Competition**, **Season**, **Rank**, **Squad**.
- ***Football Stadiums:*** https://www.kaggle.com/datasets/imtkaggleteam/football-stadiums

List of Football Stadiums across the world. I'm interested in **Stadium**, **City**, **Capacity**, **Country**, **Population**.
- ***European Football (soccer) Clubs on Google SERPs:***

https://www.kaggle.com/datasets/eliasdabbas/european-football-soccer-clubs-on-google-serps

Data about all clubs' European Tournaments wins. I'm interested in **UCL** (UEFA Champions League), **UEL** (UEFA Europa League), **CWC** (UEFA Cup Winners' Cup), **USC** (UEFA Super Cup).

Description of the work:

I want to describe matches played in the top 5 European Countries between season 2010/2011 and 2018/2019 (the intersection of the data I found). For each event (a match) I want to measure the goals of the home team, the goals of the away team and the attendance. Dimensions are the referee of the game, the date of the game, the competition in which the game was played, the stadium. The most important dimension instead is the team, both home and away. To the team I would like to associate the wins on the 4 European Tournaments, one by one. This is the only thing I found about teams in a dataset, because in reality I was searching for the creation year of each team which makes more sense from my point of view, but I couldn't find any dataset containing it and I still wanted to add something joined on the teams.

My main concern is in fact that most of the attributes are in the main table, the one of the matches, which contains the home goals, away goals, attendance, referee, stadium, date, season, competition, both the teams and the managers.

I would then join with the second dataset over the competitions to associate the type and the country to each of them. I would use the third dataset to compute the winner of each competition in the respective season by looking at the club with rank 1, as well as to store the position of each club in that season. I think of them as cross dimensional attributes. I would retrieve from the fourth dataset the city and the capacity of each stadium, then I would use the country to link the respective population to the country stored from the second one. Lastly, I would join the fifth dataset on the teams to store the number of tournaments won by each team.

If such a concern is actually a problem I would like to ask for any tips on how to overcome the problem, because I'm finding it difficult to merge some other dataset to add something big and not only details.